# JOONYOUNG JEONG

Email: 13-10113@kma.ac.kr ◇ Homepage ◇ LinkedIn

## RESEARCH INTERESTS

My research interests lie in cellular network security (5G RAN and 5GC) and computer systems security. I also enjoy dissecting binaries to uncover vulnerabilities and analyze malware functionality.

## EDUCATION

**Iowa State University** Aug 2021 – Jul 2023
M.S. in Electrical and Computer Engineering
GPA: 3.95/4.0
*Relevant coursework: Communication Systems, Advanced Communications, Network Protocol Security, Cryptography, Data Analytics in ECE, Digital Signal Processing, Introduction to Deep Learning.*

**Korea Military Academy** Mar 2009 – Mar 2013
B.E. in Mechanical Engineering

## RESEARCH EXPERIENCE

**Mobile Security Architecture for Defense 5G Infrastructure** Jul 2026 – Dec 2026
*Sponsored Research funded by ROK Army Headquarters*
*Analyzing defense mobile system architectures and domestic/international security policies (US DoD, NATO, 3GPP) to identify threat vectors relevant to 5G military network deployment.*

**Researcher, Missing Data Handling Research Team [1], [2]** Apr 2026 – Oct 2026
*Research team funded by the AI R&D Center, Korea Military Academy*
*Developed MissFlow, a flow-matching method for uncertainty-quantified tabular data imputation.*

**Cost-Aware UE-Side Mobility Management for TN-NTN Networks [3]** Sep 2025 – Aug 2026
*Independent research; single-author paper accepted to IEEE MSWiM 2026*

**Covert authentication for securing MIMO communications [4]** Jan 2022 – Jul 2023
*Supervisor: Dr. Sang Kim, Iowa State University*

**Research Assistant, Iowa State University** Aug 2022 – Jul 2023
*Summarized 5G-related papers and 5GC/RAN architectures into advisor briefs*

## PROFESSIONAL EXPERIENCE

**Rotating Faculty, Korea Military Academy** Jul 2025 – Present
*Teaching and advising cadets; participating in defense projects.*

**Planning Officer, AI R&D Center, Korea Military Academy** Jul 2026 – Present
*Research planning and coordination for AI-driven defense technology initiatives.*

**Signals Intelligence Officer, Ministry of National Defense** Aug 2023 – May 2025
*Malware Analysis and Cyber Threat Intelligence*

**Signals Intelligence Officer, Ministry of National Defense** Aug 2018 – Jul 2021
*COMINT*

**Military Intelligence Officer, Army Headquarters** Mar 2013 – Aug 2018
*General Military Intelligence*

## TEACHING

**Rotating Faculty, Korea Military Academy** Jul 2025 – Present
*Introduction to Statistics (Fall 2025, Spring 2026)*
*Introduction to Data Mining (Fall 2025)*
*Linear Algebra (Spring 2026)*

**Teaching Assistant, Iowa State University** Aug 2022 – Jul 2023
*Communication Systems II Lab (Spring 2023)*
*Communication Systems I Lab (Fall 2022)*

## SKILLS

**Programming Languages:** Python, C/C++, x86-64 Assembly, MATLAB, Simulink, R
**Analysis Tools:** Burp Suite, Ghidra, IDA Pro, x64dbg, GDB, Wireshark, AFL++

## HONORS & AWARDS

**Chairman of the Joint Chiefs of Staff Commendation** 2024
*Joint Chiefs of Staff, Republic of Korea*

**Letter of Appreciation (SES-2)** 2024
*Foreign Agency*

**Army Merit Scholarship (Full tuition and stipend)** Aug 2021 – Jul 2023
*Ministry of National Defense*

## PROJECTS

**Intrusion Detection using Denoising Autoencoder and Deep Learning** 2023
*Designed a DNN classifier using a stacked denoising autoencoder for feature learning*

## PUBLICATIONS & WORKSHOP PAPERS


[1] J. Jeong, S. Hwang, and Y. Kwon, "MissFlow: Flow matching for uncertainty-quantified imputation of missing tabular data," *IEEE Trans. Knowl. Data Eng.*, 2026, Under major revision.

[2] J. Jeong, S. Hwang, and Y. Kwon, "MissFlow: Flow matching for uncertainty-quantified imputation of missing tabular data," in *EIML@ICML 2026 Workshop*, Accepted, 2026.

[3] J. Jeong, "CALM: UE-side mobility management for integrated TN-NTN networks," in *IEEE Int. Conf. on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, Accepted; to appear, Paris, France, 2026.

[4] J. Jeong, "Covert authentication for securing MIMO communications," M.S. thesis, Iowa State University, Ames, IA, USA, 2023. [Online]. Available: `https://dr.lib.iastate.edu/bitstreams/d0ca9bbb-4436-4681-97b7-c97fa3a4ec05/download`.